1 Security Model

(placeholder for paper.tex §4)

2 Evaluation

We evaluate the main circuit and its gate contract along the research questions of the study: relation conformance (Q1), budget conservation under concurrency (Q2), revocation (Q3), transcript linkability (Q4), accountability (Q5) and cost (Q6), and compare against official implementations of neighbouring systems. The main circuit is v9.1: v9 plus an on-curve check of every encrypted chain key (Section 2.6). Cost and linkability were measured on v9; v9.1 adds 3(D+1) constraints, keeps every public input, its verifier differs only in key constants, and its proving time is not measurably different (paired runs). **Every test reported here is a finite conformance test of the implementation against the stated relation; none of it is a security proof.** Security arguments are in Section 1; the non-cryptographic invariants that are machine-checked are listed in Section 2.9.

2.1 Setup

All measurements ran on one Apple M5 Pro (15 cores, 48 GiB), node 24.14, circom 2.2.3, snarkjs 0.7.6 (Groth16 over BN254), forge/anvil 1.5.1 (Prague). The v9.1 circuit has 60,628, 105,433, 150,238 and 195,043 constraints for $D = 1..4$ (v9: 60,622–195,028) and 20 public inputs at $D = 4$ (27 before input compression). Phase 1 of the setup is the public Hermez ceremony (2^{18}) for the action, budget and revocation circuits, but a locally generated 2^{17} file for the registration circuit; *phase 2 is a single local contribution everywhere, so the keys are for evaluation only and a deployment needs a ceremony (for the registration circuit also a public phase 1).* Every experiment was pre-registered (attack lists, cases, statistics and targets committed before running); deviations, and the re-analyses added after an external audit, are reported in the per-experiment records and labelled post hoc.

2.2 Relation conformance (Q1)

Mutation testing. Every single field of a valid witness was mutated and the circuit's verdict compared with the expected one: 199, 282, 408 and 534 field mutations plus 37–76 semantic cases for $D = 1..4$, with 0 outcomes that differ from the expectation written into each case. *Differential attack suite.* An independent plaintext oracle (separate Poseidon / keccak implementations) classifies 170 attack rows on v9 (174 on v9.1) covering chain forgery, attenuation, revocation, registered roots, budget notes, the state root, the compressed accountability ciphertext, off-curve keys and proof-level retargeting. Every rejection on v9.1 records the template and line of the failing constraint, so a row cannot pass for the wrong reason. Circuit and oracle agree on 114 rejections and 28 acceptances; 7 rows fail only because the attacker supplied a wrong path for a true statement; there are 0 completeness gaps. Of the 21 rows the circuit accepts and the oracle rejects, 20 are obligations of the contract by design (context hash, required capability, clock, root windows, replay) and each is closed by a test on the v9 gate; the remaining one is the pre-registered witness-level row in which the holder legitimately proves a different context. An exhaustive single-field differential over all 534 fields at $D = 4$ finds 3 statement-level disagreements, exactly the three gate-bound inputs. At proof level, re-using a real proof with any changed action field, state root, budget nullifier or amount fails verification.

2.3 Budget conservation (Q2, H3)

We replayed concurrent spending with real transactions, all candidates of a race in one block. The v7 gate, which has permission bitmaps but no quantity semantics, accepted *all* 64 actions of 10 units (640 units against an intended budget of 100). With budget notes (v8/v9), a budget of 100 split among 32 agents (only those holding a note can build a proof) yielded exactly 100 units, and of 12 actions built on four note versions (three each) exactly 4, one per note, succeeded. All transactions of a race came from one sender in a fixed order; the nullifier set makes the outcome independent of the order. Section 2.9 gives the machine-checked invariant behind these observations.

2.4 Revocation (Q3, H2)

In a shared world with chains that share prefixes (598 checks, 0 unexpected), revoking an edge or a key blocks exactly its descendants; for the circuit itself, every leaf of the revocation tree substituted as the non-membership witness of a revoked entry is rejected. Removing the check for all hops but the last (an ablation) lets 14/14 descendants act, as it must by construction. At scale (1,230 agents), revoking 1 % and then 10 % of the delegation edges (12 and 120 edges, 84 and 301 affected agents) left 0/0 unaffected agents blocked. None of the 301 blocked agents can act: each was evaluated by the circuit with the best substitute witness for every revoked hop (the predecessor leaf and the entry's own leaf), and all 301 and 301 witnesses were rejected at the non-membership check, while 64 unaffected controls were accepted. A revocation costs 273,538 gas regardless of the size of the revocation set. The price of Δ-fresh revocation is availability: after the window, every in-flight proof built on the old root is rejected, whether or not it is affected, (64/64 and 64/64 in our samples), including those of unaffected agents, which re-prove once and succeed (32/32, 32/32).

2.5 Transcript linkability (Q4)

We generated 512 real v9 action transcripts (proof, public inputs, roots) of 64 depth-4 agents under 8 parents and 2 masters, 8 actions each. No word value is shared between transcripts (0) and no difference of ciphertext words repeats (0). Three classifiers (logistic regression, random forest, k-NN) trained on bit-level pair features, with folds grouped by agent or parent and a permutation null at the label's unit (1000 permutations, Holm), detected a link in 0 of 18 same-agent and sibling cells (largest 95 % CI lower bound of the AUC 0.471), while two planted leaks, reused ciphertext randomness and a public lineage tag, were detected in 18 of 18 cells by the same rule (Holm $p \leq 0.006$). This bounds only these attacks: a leak that moves the classifier's AUC by less than the null's 99th percentile (up to 0.70) would go unnoticed, and the master attribute cannot be tested with 8 parents (minimum attainable $p = 0.029$). Our first analysis of the same data used a rule that could not report a link; this re-analysis is post hoc.

2.6 Accountability (Q5, H4)

A 3-of-5 committee opens the accountability ciphertext with Chaum–Pedersen DLEQ partial decryptions and Lagrange interpolation in the group, without reconstructing the secret key (trusted dealer, static corruption of fewer than t members). All 65 pre-registered cases behaved as expected (0 deviations): exact opening for every 3-subset, identification of every malformed share, rejection of cross-event and wrong-epoch shares. A design comparison matters for the claim: with one ElGamal ciphertext per key (mode A), a wrong opening still produced valid curve points in 150/150 forced cases, so it cannot be detected; the KEM-plus-keystream design used by v9 (mode C) yields no valid output on a wrong key. The compressed ciphertext encrypts only the y-coordinates and the parity of x, and the auditor recovers x from the curve equation. This is correct only for keys on the curve: v9 checked the terminal key but not the intermediate ones, whose EdDSA verification uses only $8A$, and an off-curve intermediate key was accepted in all 4 positions and opened to a different key. v9.1 checks every encrypted key and rejects all 4. The compressed ciphertext is opened exactly in 10/10 fresh instances per depth, with a single key and with the threshold protocol. The adversarial opening cases ran on the v7 layout, whose key encapsulation v9 keeps.

2.7 Cost (Q6)

Latency. Fig. 1 breaks one fresh action into fetching paths, witness assembly and calculation, proving, off-chain verification, submission and inclusion. The local authorization computation takes 0.92, 1.55, 2.41 and 2.74 s for $D = 1..4$ (30 fresh actions per depth in three process blocks), of which proving takes 0.75–2.46 s; each hop

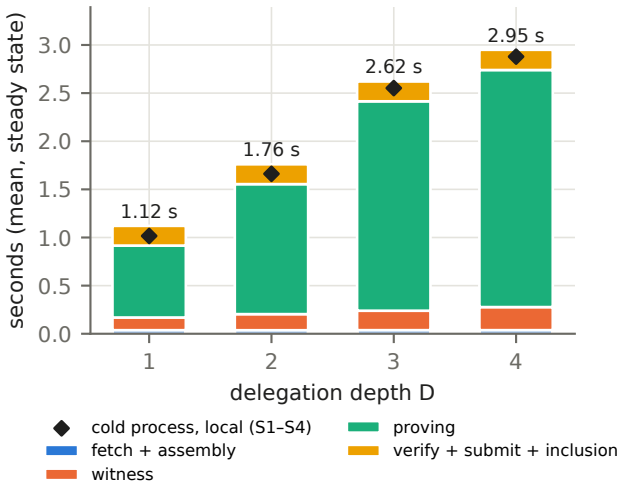


Figure 1: Lifecycle latency of one fresh action by stage (mean of 30 fresh actions per depth, steady state); diamonds: local computation in a fresh process. Proving dominates.

adds 633 ms (95 % CI 627–641). A cold process adds 100–139 ms to the action. All three pre-registered targets hold at $D = 4$: p95 local computation 2.86 s (95 % CI 2.75–3.01; cold 2.94 s) ≤ 10 s, peak memory 4.51 GiB ≤ 8 GiB, verifier-only gas 315,215 $\leq$ 500k.

Gas. The full transaction cost of an action is set by the note-tree insertion, not by the depth (Fig. 2): gas $\approx 529{,}576 + 27{,}089 \cdot \mathrm{popcount}(i) + 7{,}763 \cdot D$ ($R^2 = 0.999997$), where i is the insertion index of the change note (a post-hoc fit over indices of one tree depth; each one-bit is one more on-chain hash in the insertion). An action costs 564,147 gas on an almost empty note tree and 743,211 on average while the tree grew to 3,830 notes; a gas figure is therefore only meaningful together with the note-tree size.

Scale. 1,230 agents (1,024 at depth 4) with budgets delegated by real transfer transactions acted concurrently on the same state: 1,230/1,230 actions were accepted in 31 blocks of 30M gas (39.7 per block). One machine produced 23.1 proofs per minute.

Cost of privacy and of each component. A plaintext reference implementation of the same policy in the same contract stack (B0: ECDSA certificate chain, per-hop revocation, replay set, per-edge budget ledger; our own design, not a proven lower bound) costs 115,215–228,705 gas for $D = 1..4$ in its first execution. v9 costs 2.45–6.70× that on a fresh note tree and 3.25–8.63× at the size of our world; the range spans the accounting choices (B0's first or steady execution, with or without intrinsic gas), and the smallest values are at $D = 4$. In proving time, accountability adds 84 ms and budget notes add 63 ms at $D = 4$ (Fig. 3); at $D = 3$ accountability crosses a power-of-two FFT domain and adds 620 ms. Compressing the public inputs (v8→v9) changes proving by at most -7 ms

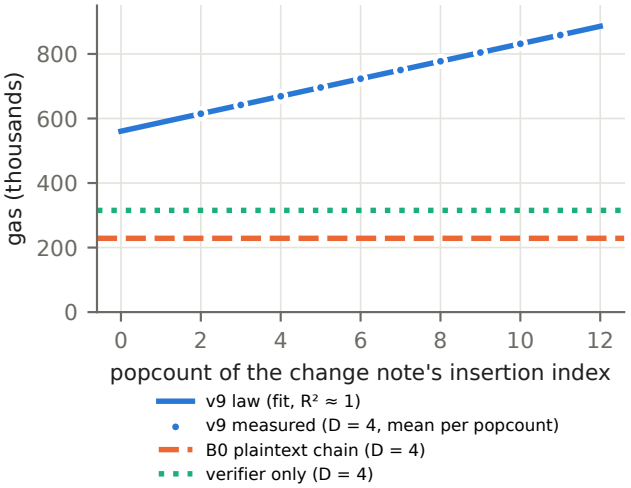


Figure 2: Execute gas at $D = 4$ against the number of one-bits of the change note's insertion index (1,024 actions), the fitted law, the plaintext reference gate B0 and the verifier alone.

(paired, $D = 2$, Holm $p = 0.031$).

2.8 Comparison with neighbouring systems

We ran the official implementations natively instead of excluding systems built on other stacks, and compared only sub-tasks they support. UCAN (go-ucan) verifies a depth-4 delegation in 0.184 ms with 1,853 bytes of tokens, with the lineage in clear and no revocation in the implementation. The DAC scheme from equivalence-class signatures shows a credential in 2.73 ms and verifies it in 16.2 ms; its code supports a single delegation level. zkcreds proves an expiry show in 260 ms (one witness, repeated) and has no delegation. ZKM, ZK-ACE and ZEBRA have no public code and are compared at protocol level: ZKM shares hidden lineage, in-circuit conservation and on-chain verification but revokes only whole subtrees and offers only voluntary disclosure; ZK-ACE has no delegation. These systems do not offer the same functionality, so the timings are not a like-for-like comparison; what the comparison shows is the combination our system adds and its cost relative to the plaintext reference above.

2.9 Machine-checked invariants

A Lean 4 model of the gate (mint, split, delegate, execute, revoke, time) proves, for every reachable state, budget conservation (unspent plus spent equals minted), subtree conservation (below any certificate edge, spending never exceeds what was minted or delegated into it), monotonic revocation and freezing (after the window nothing below a revoked edge is spent): 6 theorem statements with 0 uses of `sorry` and only Lean's standard axioms. The

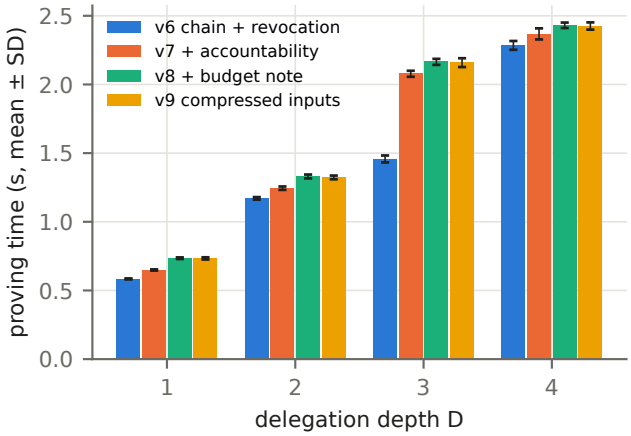


Figure 3: Proving time of successive circuit versions, 30 fresh instances per cell (mean ± SD).

model abstracts the cryptography by stated assumptions (knowledge soundness, commitment binding, nullifier injectivity) and is checked for non-vacuity: removing the nullifier check lets the model spend twice its budget.

2.10 Limitations

One machine and one prover (snarkjs); native provers were not tested. The chain is a local anvil node, so inclusion latency excludes block time and finality. Phase 2 of the setup is not a ceremony. Tree heights are fixed by the compiled circuits and were not swept; depths beyond 4 were not compiled. Linkability was tested only against the attacks above, and the anonymity and no-false-framing arguments remain paper proofs. Budget races used one sender in a fixed order; revocation at scale revoked certificates, not keys; the adversarial opening cases ran on the v7 ciphertext layout.